

Name: _____

Solving Radical Equations

Day 1 Practice: Radical Already Isolated

Problems 1-8 Solve the radical equation. Check your solution.

1. $\sqrt{x+2} = 4$

2. $\sqrt{3x-5} = 4$

3. $\sqrt{x-7} = 6$

4. $\sqrt{5x+1} = 6$

5. $\sqrt{2x+9} = 5$

6. $\sqrt{x+10} = 8$

7. $\sqrt{4x-3} = 5$

8. $(6x+4)^{\frac{1}{2}} = 10$

9. $(x+7)^{1/2} = 5$

10. $(2x-3)^{1/2} = 7$

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Day 2 Practice: Isolating the Radical & No Solution

Problems 1-8 Solve the radical equation. Check your solution. Write “No Solution” if applicable.

1. $\sqrt{x+3} + 2 = 6$

2. $\sqrt{2x+5} - 1 = 4$

3. $\sqrt{x-2} + 7 = 3$

4. $\sqrt{3x+1} - 5 = 1$

5. $3(x+2)^{\frac{1}{2}} = 12$

6. $\sqrt{x+8} + 9 = 5$

7. $(5x-4)^{\frac{1}{2}} - 2 = 5$

8. $4(x-1)^{\frac{1}{2}} = 8$

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Day 3 Practice: Two Radicals & Applications

Problems 1-6 Solve the radical equation. Check your solution.

1. $\sqrt{2x+5} = \sqrt{x+8}$

2. $\sqrt{4x-1} = \sqrt{x+8}$

3. $\sqrt{3x+10} = \sqrt{5x-6}$

4. $\sqrt{6x-5} = \sqrt{4x+3}$

5. $\sqrt{2x+15} = (5x)^{\frac{1}{2}}$

6. $(7x-2)^{\frac{1}{2}} = (3x+14)^{\frac{1}{2}}$

Problems 7-8 Solve the application problem.

7. The time t (in seconds) it takes an object to fall from a height h (in feet) is given by:

$$t = \sqrt{\frac{h}{16}}$$

A ball takes 3 seconds to hit the ground. From what height was it dropped?

8. The speed s (in miles per hour) of a car that leaves skid marks d feet long can be estimated using the formula:

$$s = \sqrt{24d}$$

A car was traveling 60 mph. How long were its skid marks?